Bohdan Mykhats

Senior Full-Stack Engineer | System Architect

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SUMMARY

Senior Full-Stack Engineer and System Architect with 5+ years of software engineering experience. Focused on architecting high-performance web applications, scalable distributed backend systems, and interactive 3D graphics engines. Proven track record of leading development teams, translating business requirements into technical architectures, and delivering robust, cloud-native solutions.

TECHNICAL ARSENAL

System Architecture

Domain-Driven Design (DDD),
Clean Architecture,
CQRS,
Microservices, Modular Monolith, RESTful API Design

Frontend

Angular,
React,
NextJS, LitJS, SolidJS

Backend

.NET,
NodeJS,
NestJS

Cloud & CI/CD

AWS (ECS, EC2, Lambda, S3), **Azure**,
GitHub Actions, Azure Pipelines, Docker, Docker Compose

Databases

Supabase (PostgreSQL), PostgreSQL, MSSQL, Redis, MongoDB

WebGL & 3D

ThreeJS, WebGPU, VR, Custom Shaders, Linear Algebra

PROFESSIONAL EXPERIENCE & PROJECTS

Maplyx
Tech Lead / System Architect

Architected and engineered a highly scalable EdTech SaaS featuring a custom linear-spined mind map editor for sequential course building with real-time collaboration. Collaborated directly with stakeholders to gather requirements, designed the high-level system architecture, and delegated implementation tasks to the development team.

.NET 10 · Angular · Supabase · Centrifugo

3D Property Editor & Walkthrough
Tech Lead / System Architect

Designed and deployed a highly scalable infrastructure utilizing **AWS EC2, ECS, Lambda** (HTTP endpoints), **Express.js** with **Socket.io** for real-time connections, **KMS, S3, and DynamoDB**. The system currently scales seamlessly in production and supports two independent frontend applications. Achieved massive VRAM reductions using custom texture atlases and instanced meshes for a cloud-connected VR walkthrough presentation system. Collaborated directly with stakeholders to gather requirements, designed the high-level system architecture, and delegated implementation tasks to the development team.

WebGL · Three.js · AWS Lambda · Node.js · Socket.io · AWS EC2 · AWS ECS · DynamoDB · KMS

Employee Management App 🔒 NDA
Frontend Engineer

Engineered cross-platform desktop applications (Windows/macOS) focusing on native component development for optimized performance. Actively managed technical backlogs, delegated tasks, and communicated cross-team dependencies to ensure smooth delivery.

React Native for Desktop · Native Modules · Swift · Objective-C · WPF

Medical App 🔒 NDA
Full-Stack Developer

Maintained a secure, scalable healthcare platform with integrated payment processing and robust MSSQL database architecture.

React · .NET Framework · MSSQL · Payment Gateways · Quartz · SignalR

CUSTOM 3D ENGINE DEVELOPMENT

Custom Geometry Engine
Lead Developer

Engineered a custom geometry engine ("mini-blender") from scratch. Implemented dynamic geometry generation (primitive/custom shapes) and complex topological operations including extrude/intrude, face inset, curve-based segment cuts, and dome extrude. Developed robust systems for automated UV and normal calculations, material group management for multi-texturing, and integrated a Command Pattern for comprehensive undo/redo functionality.

WebGL · Three.js · Design Patterns · Linear Algebra

R&D & HIGH-PERFORMANCE ENGINEERING

A* Pathfinding Implementation

Engineered a high-performance A* pathfinding algorithm utilizing Unity's DOTS (Data-Oriented Technology Stack) and Burst Compiler. Leveraged advanced parallelization techniques to achieve a 37.5x performance speedup.

Hardware-Accelerated Signal Processing (.NET)

Developed a .NET application for hardware-accelerated signal and image processing. Implemented core features including FFT, STFT, wavelets, image operations (kernel), and affine transformations. Engineered a web interface for task selection (audio/image) with dynamic routing to CPU, DirectX, or OpenCL parallelization, actively tracking CPU/RAM/GPU resource consumption.

</> OPEN SOURCE CONTRIBUTIONS

.NET Feature Management

Developed and published an MIT-licensed feature management package specifically designed for .NET Minimal APIs, providing robust and flexible feature toggling capabilities.

.NET Result Pattern Handler

Created an MIT-licensed generic "Result Pattern" package offering a unified handler for success and error responses, configurable at both the global application level and for specific endpoints.

EDUCATION

Master's Degree in Software Engineering
Uzhhorod National University, 2025

Bachelor's Degree in Computer Engineering
Uzhhorod National University, 2024

Sigma Software University — .NET Camp Graduate

LANGUAGES

English: Upper-Intermediate (B2)